

I. Project Title

EcoSort: A Smart Waste Segregation Assistant and Logger

II. Problem Statement

A great many students and members of households have difficulty in properly dividing their rubbish into the categories of biodegradable, non-biodegradable, and recyclable materials. The reason for this is that people do not know which category a particular item should go into, particularly when it comes to less common items such as sanitary products, foil wrappers, or packaging made from a combination of materials. Consequently, waste is frequently placed in the incorrect bin, thereby contaminating the recyclable materials, increasing the amount of waste going to landfills, and making composting and recycling schemes less effective. The issue impacts students, teachers, canteen staff, and the broader community, and it is important to address it since incorrect waste separation leads to pollution, clogs waterways, and results in inefficient waste management in both schools and barangays.

III. Project Objectives

- 1 Develop a program that correctly classifies at least 90% of a test list of 30 common waste items into Biodegradable, Non-biodegradable, or Recyclable within the first month of development.
- 2 Reduce the average time it takes a user to identify the correct waste category for an item to under 15 seconds per entry, measured through simple user testing before the end of the quarter.
- 3 Have at least 20 classmates test the initial version of the program and log at least 5 waste items each by the end of the trial period, to confirm the tool is usable and useful.

IV. Planned Features

- Waste Item Lookup - user types or selects a waste item, and the system returns its correct category.
- Waste Category Reference List - a built-in list of common items already classified for quick lookup. Daily
- Waste Log - records each item a user disposes of, along with the date and category.
- Category Summary Dashboard - shows a simple count/percentage of items logged per category over time.
- Input Validation - rejects blank entries and flags items not found in the reference list instead of guessing.

V. Planned Inputs and Outputs

Inputs	Outputs
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Name/keyword of the waste item	The correct waste category (Biodegradable, Non-biodegradable, or Recyclable)
Date the item was disposed of	A confirmation message that the item was logged successfully
User's choice to view their log or summary	A summary report showing total items logged per category
User's own guess of the category, for a quick quiz mode	Feedback on whether the user's guess was correct, with the right answer shown

VI. Logic Plan (Pseudocode)

START

DISPLAY menu: [1] Look up an item, [2] Log a waste item, [3] View summary, [4] Exit GET user choice

IF choice = 1 THEN

GET item_name FROM user

IF item_name is blank THEN

DISPLAY "Please enter an item name."

ELSE

SEARCH item_name IN reference_list

IF found THEN

DISPLAY category of item_name

ELSE

DISPLAY "Item not found. Try a different keyword." END IF

END IF

ELSE IF choice = 2 THEN

GET item_name, date FROM user

IF item_name is blank OR date is blank THEN

DISPLAY "Item name and date are required."

ELSE

SEARCH item_name IN reference_list

IF found THEN

ADD (item_name, category, date) TO waste_log

DISPLAY "Item logged successfully."

ELSE

DISPLAY "Item not found. Cannot log an unknown item." END IF

END IF

ELSE IF choice = 3 THEN

COUNT items in waste_log per category

DISPLAY summary (counts and percentages per category)

ELSE IF choice = 4 THEN

END PROGRAM

ELSE

DISPLAY "Invalid choice. Please select 1-4." END IF

REPEAT until choice = 4 END

Submission via Google Classroom

GitHub Repository Link: <https://github.com/KharlTheo/ecosort>